

Huaibing (Lorraine) Yuan

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RESEARCH OVERVIEW

My research focuses on responsible AI governance, i.e., examines how data-driven technologies and generative AI reshape operational systems across organizational workflows and production networks. I develop algorithmic tools and analytical models to study how privacy, incentives, and workflow design determine whether these technologies improve decision quality and operational performance. In a complementary empirical stream, I apply transformer models to large-scale trip and charging datasets for electric vehicles (EVs) to evaluate its operational and societal impacts.

ACADEMIC EMPLOYMENT

Cornell Tech Postdoctoral Fellow	2026–Present
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EDUCATION

Cornell University Ph.D. in Operations, Technology & Information Management Samuel Curtis Johnson Graduate School of Management	2026
Cornell Tech M.Eng. in Operations Research & Information Engineering	2020
Renmin University of China B.S. in Mathematics and Applied Mathematics B.A. in Economics	2019

RESEARCH INTERESTS

Topics: Privacy for AI, Human-AI Orchestration in Operations, Sustainable Transportation & Mobility.

Methodologies: Data-Driven Optimization, Economic and Statistical Modeling, Mechanism Design, Transformer Models, Empirical OM.

WORKING PAPERS

[1] “Privacy-Preserving Data-Driven Decisions with Strategic Privacy Design.” With Elena Belavina.
Under Review after a **Major Revision** at *Management Science*.

- Third Place, YinzOR Conference Competition
- Accepted to NYC Privacy Day at Google

Summary: Studies how firms can produce operational knowledge from sensitive data while limiting privacy leakage, showing that machine learning algorithms differ in their robustness to information exposure; develops algorithmic solutions that balance data-sharing benefits with privacy risks.

Methodologies: Data-Driven Optimization, Empirical Calibration, Differential Privacy.

- [2] “Governing Agentic AI Workflows: Oversight with Agentic and Assistive Designs.” With Karan Girotra.
In preparation for submission to *Management Science*.

– Accepted to Fisher AI in Business Conference’ 26

Summary: Develops a stochastic principal-agent model of multi-step AI workflows and characterizes how firms should assign human and AI roles when oversight is costly, hidden, and AI performance is jagged.

Methodologies: Stochastic Modelling, Principal-Agent Theory, Mechanism Design.

- [3] “Data-Driven Decisions in Supply Chains: Contracts, Algorithms, and Efficiency.” With Elena Belavina, Karan Girotra and Xiaoyue Yan.

In preparation for submission to *Management Science*.

Summary: Studies how learning algorithms reshape coordination in supply-chain networks, showing that algorithmic bias and variance can reverse standard contract-efficiency results across prevalent data-driven methods.

Methodologies: Data-Driven Algorithms, Contract Theory.

- [4] “From Mobility Trajectories to EV Charging Decisions: Understanding EV Charging Behavior via Language-Model Paradigms.” With Yu Nu.

Working paper, Submitted to *NeurIPS ML×OR Workshop*.

Summary: Collaborates with a large-scale EV data platform to evaluate drivers’ charging decisions as a sequence of mobility and charging events using a language-model paradigm, identifying key drivers of range anxiety and the precautionary battery buffers. Conducts counterfactual analysis to design charging guidance that eases public burdens while improving driver utility.

Methodologies: Language-Model Paradigm, Empirical Analysis, Structure Estimation.

ONGOING PROJECTS

- [5] “Data-Driven Newsvendor” (MBA Teaching Case). With Karan Girotra.

- [6] “Privacy-Preserving Collaborative Learning.”

- [7] “AI Token Allocation in a Production Network.”

TEACHING

Johnson Graduate School of Management, Cornell University

Instructor

Guest Lecturer, AI and Digital Transformations (MBA Elective)

Fall 2026

Presented on structuring human oversight of multi-step AI agent workflows.

Co-instructor, Managing Operations (MBA Core)

2022, 2023

Evaluations: Spring 2023: 4.85/5 (26 respondents); Fall 2022: 4.75/5 (20 respondents)

Teaching Assistant

Business Data Analysis with SQL (MBA Elective)	Summer 2024
Generative AI for Business Transformation (eCornell Certificate Program)	Spring 2024
Managing Operations (EMBA)	Fall 2023, Spring 2021
Operations Management (Cornell-Peking MBA)	Spring 2023
Innovation & New Venture Creation/The Customer Experience (MBA Elective)	Fall 2021

<i>Directed Reading Mentor</i>	Spring 2023
Guided MBA students in translating an academic paper's algorithm into an applied industry showcase.	

Independent Instruction, Paths Academics

Designed and taught applied curriculum in *Python for Data Science* and *Comparative Economic Systems* for over 200 undergraduate students; culminated in student projects and portfolios.

SELECTED PRESENTATIONS

Privacy-Preserving Data-Driven Inventory Management

INFORMS Annual Conference, Atlanta, GA	Oct 2025
INFORMS Workshop on Data Science, Atlanta, GA	Oct 2025
YinzOR Student Conference, Pittsburgh, PA	Aug 2025
MSOM Annual Conference, London, UK	Jun 2025
Digital Life Initiative Workshop, New York, NY	Apr 2025
NYC Operations Day (Poster Session), New York, NY	Mar 2025
NYC Privacy Day at Google, New York, NY	Oct 2024

Governing Agentic AI Workflows: Oversight with Agentic and Assistive Designs

Fisher AI in Business Conference, Columbus, OH	Oct 2026 (scheduled)
INFORMS Annual Conference, San Francisco, CA	Nov 2026 (scheduled)
– Nov 2nd, 11:00 AM–12:15 PM, Moscone South-204	
– Nov 3rd, 3:45 PM–4:00 PM, Moscone South-70	

Understanding EV Charging Behavior via Language-Model Paradigms

POMS Annual Conference, Orlando, FL	Apr 2027 (scheduled)
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AWARDS AND HONORS

Byron E. Grote Johnson Professional Scholarship, Cornell University	2021, 2024
Digital Life Initiative Doctoral Fellowship, Cornell University	2024
Cornell University Doctoral Fellowship	2020–2026
Noel Croucher Graduate Fellowship (\$20,000), Cornell University	2019
Lester B. Knight Scholarship (\$40,000), Cornell University	2019
First Class Study Scholarship, Renmin University of China	2018

PROFESSIONAL EXPERIENCE

Uber	Sep 2019–Dec 2019
<i>Student Consultant, Uber Eats Marketplace Intelligence</i>	New York, NY
Collaborated with the Uber Eats Marketplace Intelligence team to design a pooled delivery dispatch system aimed at	

improving grocery access in food deserts. Developed a scheduling algorithm to batch customer orders and proposed routing policies for food trucks to enhance delivery efficiency and coverage.

Cardinal Operations

Algorithm Engineer Intern

Sep 2018–Apr 2019

Beijing, China

Conducted sales forecasting using machine learning algorithms and derived pricing solutions for client companies at an enterprise-scale big data optimization solutions provider.

ACADEMIC SERVICE

Reviewer, MSOM Annual Conference	2025
Doctoral Fellow, Digital Life Initiative, Cornell University	2024–2025
Co-chair, OTIM PhD Forum, Cornell University	2022–2023
Session Co-chair, INFORMS Annual Conference	2022

REFERENCES

Prof. Elena Belavina

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Prof. Yao Cui

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